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Map Manager

 Map Matching
Data
Requirements
Version [1.1]



Department
of Transport
and Planning



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1. Introduction

The purpose of this document is to outline clear expectations for the quality of DTP-OSM that is the output of the Map Matching process delivered by Map Manager.

Map Manager was developed in 2023 to support the Optimal Reality Suite of tools, particularly LRS, RID, OneView and RCIS. At a high level, it aims to deliver the following functionality:

1. Maintenance of **DTP-OSM dataset**, including alternate weekly ingestion of Vicmap Transport (TR_Roads) and OpenStreetMap (OSM) to maintain currency of DTP-OSM.
2. Development of an interface to review and manually edit DTP-OSM.

This document is only concerned with point 1 in the list above.

Currently, Map Manager does not produce a DTP-OSM dataset of a quality and completeness that meets DTP needs. As such this document aims to make explicit the quality metrics required to consider the Map Matching process suitable for use to support DTP applications.

The document also provides a short summary of the designed outcomes of Map Matching, and how data quality can be measured.

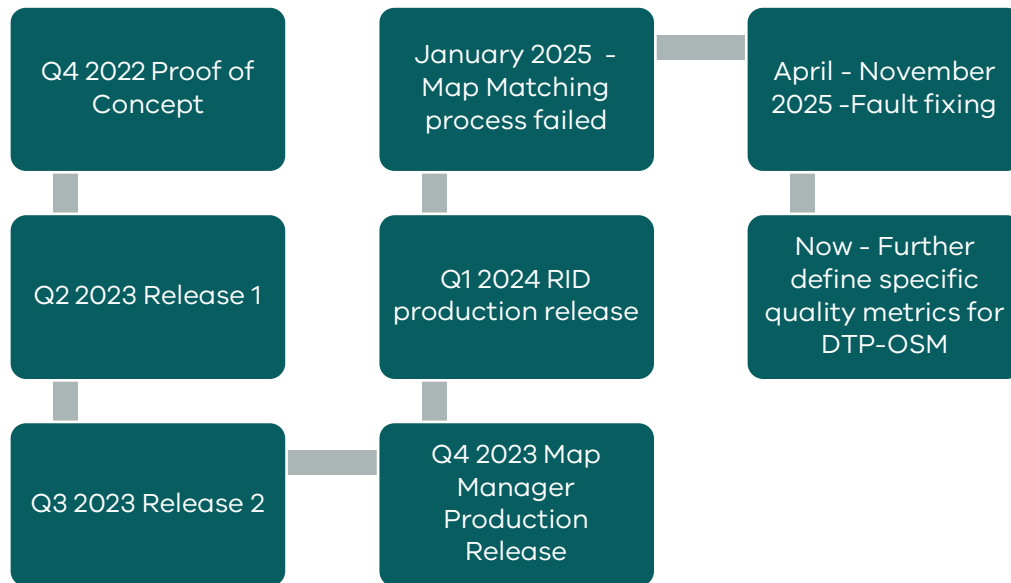
2. Map Matching and DTP-OSM

Map Matching within Map Manager was designed to have the following functionality:

1. Automated maintenance
 - a. Extractions from Vicmap and OSM
 - b. Road matching and attribute assignment
 - c. Exception reporting
2. High Update Frequency (weekly alternating between Vicmap and OSM)
3. Detailed spatial representation of roads, breaking at both in line with OSM and TR_Road segmentation.
4. Supports a routable road network
5. Persistence of DTP custom (DTP operational), attributes including Road Numbers, Road Maintenance Category, etc.
6. Enrichment with authoritative Vicmap Road names and other attributes.

A summary of the Map Manager Timeline is shown in Figure 1.

Figure 1 Summary timeline



Extensive testing of Map Manager functionality and interface was undertaken prior to acceptance. Data quality was deemed adequate, without qualitative testing or extensively defined metrics. Case study errors were identified and used to highlight future changes required. Dataset wide quality metrics were not adequately defined or reported. In the project closure report, dated September 2024, data quality of DTP-OSM was identified as a residual issue.

2.1. Existing Quality Reporting

An example of the how the outputs of Map Matching have been reported by the provider are shown in Figure 2. It can be quickly seen that although this information provides several metrics, the reviewer is not able to understand whether the numbers represent poor matching or true change in the networks. As such, this document attempts to define more informative data quality metrics and hence seeks to allow for explicit and specific requirements to address the data quality issue.

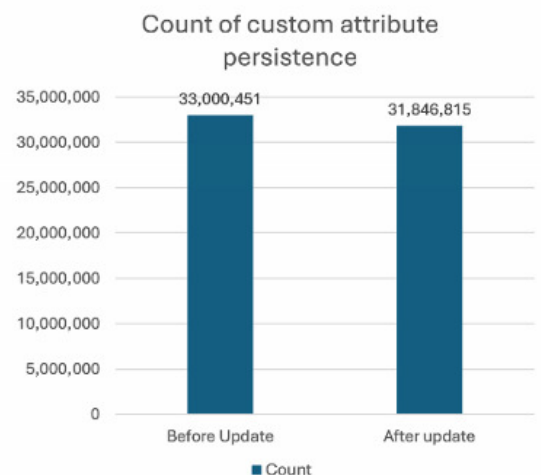
Figure 2 Sample Attribute Matching Report

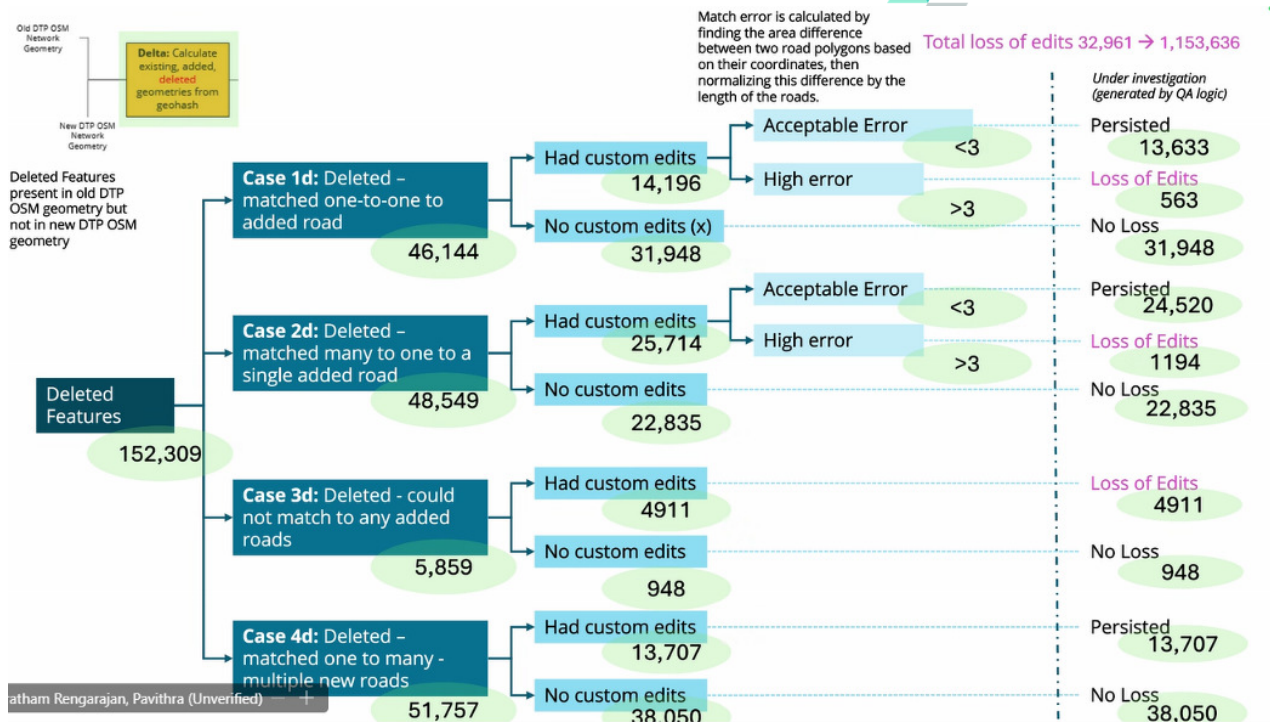
Custom Attribute Persistence – Database Analysis

- **Total custom attributes marked b_is_latest before source update:** 33,000,451
- **Total custom attributes marked b_is_latest after source update:** 31,846,815
- **Custom attributes lost during update:** 1,153,636
- **Loss percentage:** ~3.5% (Calculated as $(1,153,636 / 33,000,451) \times 100$)

Interpretation:

- Each feature has ~35 custom attributes in mm_custom_attribute table
- Therefore, ~32,961 features $(1,153,636 \div 35)$ **no longer have persisted custom attributes** in the current network file





3. Data Quality

This section of the document briefly introduces the key Victorian Data Quality Dimensions to align language and reporting.

As defined, data quality concerns the accuracy and trustworthiness of spatial data and its fitness for the defined use. Key dimensions for assessment according to the Victorian Data Quality Guideline¹ are shown in Figure 3.

Data Quality issues can be caused in each of these dimensions, related to people, processes, technology or a combination of these.

Figure 3 Data Quality Dimensions



The following section summarises proposed metrics under relevant dimensions.

¹ Data policies and standards | vic.gov.au

4. Data Quality Metrics

Key to developing metrics for each of the Quality Element in the diagram is the **Fit for Purpose** dimension.

As the defined DTP-OSM was designed to be a spatial source of truth for Victoria's road network. It should thus:

- a) represent the geometry of the roads of Victoria, (plus a 50km buffer into NSW and South Australia) to allow for the location and storage of events and assets with location information provided by the LRS
- b) store and persist custom attributes of Declared Roads and from Vicmap to subsequent versions of DTP OSM if the geometry has not changed
- c) store critical attributes to a timestamped version of DTP OSM.

Table 1 summarises the quality metrics that are further detailed in Section 5 of this report.

Table 1 Summary Proposed Quality Metrics

Property	Description	Metric	Note
Consistency			
Comparison with source datasets	DTP-OSM will include the geometry of all roads from the current version of OSM	98% of road segments in OSM will be present in DTP OSM	
	DTP-OSM will include road names of all priority roads from the current version of Vicmap	99% of unique Road Names on priority roads in Vicmap are present in DTP-OSM	
Consistency over time	DTP-OSM should persist custom attributes from the previous version where features have not changed geometry	100% of features with unchanged geometry must retain their custom attributes from the previous version	
	DTP-OSM should persist custom attributes from the previous version where features have changed geometry	At least 95% of those features must retain their custom attributes from the previous version, where the attribute remains logically applicable	This does not apply to new features.
	When running test datasets, map matching should produce identical outputs	0% of attributes or features should change between runs	
Completeness			
Attributes	DTP-OSM will have no missing attributes from defined fields	All required attributes must be present for every feature	See Appendix for Fields
Network	DTP-OSM will be a fully connected and topologically sound road network where all segments integrate correctly, without gaps or breaks, and comply with required data integrity rules.	The resulting dataset must form a complete, connected network with no unintended gaps or breaks	
Timeliness / Currency			
Currency	Map matching allows for DTP-OSM to include changes in OSM and Vicmap that are no more than 2 weeks old	Map Matching algorithm should take < 24 hours to run AND DTP-OSM must incorporate all changes that are no more than two weeks old	

Fit for Purpose			
LRS	DTP-OSM has required attributes and geometry for use in LRS	all attributes required for LRS use must be present	
Intersections	For locations that have 2 or more priority roads with different names intersecting an intersection point will be created	DTP-OSM must include an intersection point at 100% of relevant locations	In line with existing intersection creation rules
Accuracy			
Attribute Matching	Attributes should be matched to correct road segments	<1% of features should have attributes incorrectly matched from intersecting roads	
		0% of features should have attributes incorrectly matched from features of different types (class codes)	For example, a laneway should not be incorrectly matched with a freeway
	Roads should have consistent attributes along the route, with no gaps	Routes should be able to be built for all DTP roads, with <2 features created for each route	

5. Proposed Quality Metrics

This section defines the complete set of functional requirements and acceptance criteria that will be used to assess whether the DTP-OSM solution meets business, technical and data quality expectations. The scenarios are written in Gherkin format and represent the specific behaviours and outcomes that will be tested during validation; they are not illustrative examples. Collectively, they describe the full suite of checks that will be applied to confirm the solution produces a fit-for-purpose, accurate, complete, connected and maintainable road network and associated services. Compliance with all scenarios in this section is required for the solution to be considered acceptable.

5.1. Consistency

Description: When updating DTP-OSM from OSM all new feature geometry will be brought across DTP-OSM will include the geometry of all roads from the current version of OSM

Feature 2: Consistency of Road Geometry in DTP-OSM

Ensure that the geometry of all Priority roads* in DTP-OSM is consistent with the current OSM dataset.

Scenario 1: Verify road segment geometries

Given the DTP-OSM dataset of Priority road* segments

And the OSM dataset as a reference

When the geometry of each DTP-OSM road segment is compared to OSM

Then at least 98% of Priority road* segments must have geometries consistent with OSM

And consistent geometry is defined as segment vertices representing roads will be within 0.01m of the OSM feature geometry.

**Note: Additional nodes created from segmentation should be excluded from this analysis.*

Scenario 2: Reporting of geometry consistency

Given the geometry comparison has been completed

When the vendor delivers the DTP-OSM dataset

Then a report must be provided showing:

- Total number of Priority road* segments
- Number and percentage of segments with geometries consistent with OSM
- Any segments with inconsistent geometry, with potential reasons
- The report must be delivered within 24 hours in a structured, machine-readable format, and must include DTP-OSM unique feature IDs, OSM unique feature IDs, and, where applicable, Vicmap feature IDs.

Scenario 3: Handle geometry inconsistencies

Given geometry consistency is below 98%

When the report is reviewed

Then the vendor must identify segments with inconsistent geometries

And provide a brief explanation for the cause

And propose corrective actions for review by DTP's geospatial team

Note: Priority road is defined in Appendix 1*

Description: DTP-OSM will include road names of all Priority roads* from the current version of Vicmap

Feature 3: Verification of Road Name Attributes in DTP-OSM

Ensure that DTP-OSM includes road name attributes for all Priority road* from the current Vicmap dataset, with at least 99% coverage.

Scenario 1: Verify inclusion of road name attributes

Given the current Vicmap dataset of Priority road* segments

And the DTP-OSM dataset generated from it



When the road name attributes in DTP-OSM are reviewed

Then at least **99%** of unique road names in Vicmap must be present in DTP-OSM

Scenario 2: Reporting of road name coverage

Given the review of road name attributes has been completed

When the vendor delivers the DTP-OSM dataset

Then a report must be provided showing:

- Total number of unique road names in Vicmap
- Number and percentage of road names represented in DTP-OSM
- Any missing road names, with potential reasons
- The report must be delivered within 24 hours in a structured, machine-readable format, and must include DTP-OSM unique feature IDs, OSM unique feature IDs, and, where applicable, Vicmap feature IDs.

Note: Priority road is defined in Appendix 1*

Description: For Locations that have 2 or more priority roads with different names intersecting an intersection point will be created

Feature 4: Comparison of Road Intersection Points with Source Datasets

Ensure that DTP-OSM correctly creates intersection points where two or more roads with different names intersect, consistent with the source datasets.

Scenario 1: Verify creation of intersection points

Given the current Vicmap and OSM datasets containing Priority roads* segments

And the DTP-OSM dataset generated from them

When locations are analysed for intersections of two or more roads with different names

Then DTP-OSM must include an intersection point at 100% of those locations

And “unnamed” roads should not trigger this intersection creation

Scenario 2: Reporting of intersection point comparison

Given the intersection point comparison has been completed

When the vendor delivers the DTP-OSM dataset

Then a report must be provided showing:

- Any missing intersection points, with potential reasons
- The report must be delivered within 24 hours in a structured, machine-readable format, and must include DTP-OSM unique feature IDs, OSM unique feature IDs, and, where applicable, Vicmap feature IDs

Scenario 3: Handle missing intersection points

Given intersection point creation is below 100%

When the report is reviewed

Then the vendor must identify locations where intersection points are missing

And provide a brief explanation for the cause

And propose corrective actions for review by DTP’s geospatial team

Note: Priority roads are defined in Appendix 1*

Description: DTP-OSM should persist custom attributes from the previous version where features have not changed geometry

Feature 5: Consistency of Custom Attributes Over Time in DTP-OSM

Ensure that custom attributes from the previous version of DTP-OSM are retained for features whose geometry has not changed.



Scenario 1: Verify persistence of custom attributes

Given the previous version of the DTP-OSM dataset

And the current version of the DTP-OSM dataset

When features are compared between versions

Then 100% of features with unchanged geometry must retain their custom attributes from the previous version

Scenario 2: Reporting of attribute persistence

Given the comparison between DTP-OSM versions has been completed

When the vendor delivers the updated dataset

Then a report must be provided showing:

- Total number of features with unchanged geometry
- Number and percentage of those features that retained their custom attributes
- Any features where attributes were not retained, with potential reasons
- The report must be delivered within 24 hours in a structured, machine-readable format, and must include DTP-OSM unique feature IDs, OSM unique feature IDs, and, where applicable, Vicmap feature IDs.

Scenario 3: Handle attribute persistence below target

Given custom attribute persistence is below 100%

When the report is reviewed

Then the vendor must identify features where attributes were not retained

And provide a brief explanation for the cause

And propose corrective actions for review by DTP's geospatial team

Description: DTP-OSM should persist custom attributes from the previous version where features have changed geometry

Feature 6: Consistency of Custom Attributes for Changed Features Over Time in DTP-OSM

Ensure that custom attributes from the previous version of DTP-OSM are retained, where possible, for features whose geometry has changed.

Scenario 1: Verify persistence of custom attributes for changed features

Given the previous version of the DTP-OSM dataset

And the current version of the DTP-OSM dataset

When features with changed geometry are identified

Then at least 95% of those features must retain their custom attributes from the previous version, where the attribute remains logically applicable

And consistent geometry is defined as segment vertices representing roads will be within 5.00 metres of the OSM feature geometry.

Note that additional nodes created from segmentation should be excluded from this analysis

Scenario 2: Reporting of attribute persistence for changed features

Given the comparison between DTP-OSM versions has been completed

When the vendor delivers the updated dataset

Then a report must be provided showing:

- Total number of features with changed geometry
- Number and percentage of those features that retained their custom attributes
- Any features where attributes were not retained, with potential reasons
- The report must be delivered within 24 hours in a structured, machine-readable format, and must include DTP-OSM unique feature IDs, OSM unique feature IDs, and, where applicable, Vicmap feature IDs.

Scenario 3: Handle attribute persistence below target

Given custom attribute persistence for changed features is below 95%

When the report is reviewed

Then the vendor must identify features where attributes were not retained



And provide a brief explanation for the cause
And propose corrective actions for review by DTP's geospatial team

Description: When running test datasets, map matching should produce identical outputs

Feature 7: Deterministic Map Matching Output in DTP-OSM

Ensure that running map matching on test datasets produces identical outputs across runs.

Scenario 1: Verify identical outputs for repeated test runs

Given a test dataset in DTP-OSM

And the map matching process is executed multiple times on the same dataset

When outputs from each run are compared

Then 0% of attributes or features should change between runs

Scenario 2: Reporting of consistency across test runs

Given multiple map matching runs have been completed

When the vendor delivers the test outputs

Then a report must be provided showing:

- Total number of features in the test dataset
- Number and percentage of features and attributes that remained identical across runs
- Any changes detected, with potential reasons
- The report must be delivered within 24 hours in a structured, machine-readable format, and must include DTP-OSM unique feature IDs, OSM unique feature IDs, and, where applicable, Vicmap feature IDs.

Scenario 3: Handle unexpected changes in test outputs

Given any feature or attribute changes are detected between runs

When the report is reviewed

Then the vendor must identify the features or attributes that changed

And provide a brief explanation for the cause

And propose corrective actions for review by DTP's geospatial team

5.2. Completeness

Description: Ensure the system generates a fully connected and topologically sound road network where all segments integrate correctly, without gaps or breaks, and comply with required data integrity rules.

Feature 1: Produce a Complete and Connected Road Network

Ensure that DTP-OSM delivers a fully connected, topologically valid, and complete network that supports downstream systems, location referencing, routing, validation processes, and manual editing workflows.

Scenario 1: Network is fully connected with no breaks or gaps

Given the map matching process integrates OSM, Vicmap and DTP-managed data

And all required fields, domains, defaults and class codes have been validated

And attribute and geometry persistence rules have been applied

And custom attributes are retained where features are unchanged

And topology checks confirm correct spatial relationships

When the DTP-OSM network is generated

Then the resulting dataset must form a complete, connected network with no unintended gaps or breaks

And all road segments must connect where their geometry and attributes indicate they should

And no isolated segments may exist unless explicitly defined as standalone features

And all DTP managed roads must be present with required attributes populated

And the network must support location referencing operations, including routing and linear referencing across multiple connected



segments

Scenario 2: Connectivity maintained across updates

Given a previous version of DTP-OSM is available

And the new output incorporates updated OSM and Vicmap data no more than two weeks old

And unchanged features retain their custom and core attributes

And changed features retain attributes at the required persistence rate

When the new version is compared with the previous

Then network connectivity must be equal to or improved relative to the previous version

And no connectivity regressions may be introduced by the map matching process

And any required manual edits must be completed within the allowed effort limits

And exceptions must be recorded in the standard reporting format.

Description: DTP-OSM will have no missing attributes from the fields attached in Appendix 1

Feature 8: Completeness of Required Attributes in DTP-OSM

Ensure that all required attributes listed in Appendix 1 are present for every feature in the DTP-OSM dataset.

Scenario 1: Verify presence of required attributes

Given the DTP-OSM dataset has been generated

And the list of required attributes is defined in Appendix 1

When each feature in DTP-OSM is reviewed

Then all required attributes from Appendix 2 must be present for every feature

And no required attribute may be missing

Scenario 2: Reporting of attribute completeness

Given the completeness review has been completed

When the vendor delivers the DTP-OSM dataset

Then a report must be provided showing:

- Total number of features reviewed
- Any features missing one or more required attributes
- The names of missing attributes and potential reasons
- The report must be delivered within 24 hours in a structured, machine-readable format, and must include DTP-OSM unique feature IDs, OSM unique feature IDs, and, where applicable, Vicmap feature IDs.

Scenario 3: Handle missing attributes

Given any required attributes are missing from any feature

When the report is reviewed

Then the vendor must identify the cause of the missing attributes

And propose corrective actions for review by DTP's geospatial team

5.3. Timeliness / Currency

Description: Map Matching reports exceptions for data managers to check

Feature 9: Completeness of Map Matching Exception Reports in DTP-OSM

Ensure that map matching generates exception reports so data managers can review and act on gaps or issues.

Scenario 1: Verify generation of exception reports

Given DTP-OSM is processing the road network

When map matching encounters incomplete, conflicting, or uncertain data

Then exception reports must be generated for data managers to review

And all relevant exception details must be included in the reports



Scenario 2: Reporting of exceptions

Given map matching has been completed

When the vendor delivers the DTP-OSM dataset

Then a report must be provided showing:

- Total number of exceptions identified
- Types of exceptions and affected features
- Any patterns or clusters of data issues
- The report must be delivered within 24 hours in a structured, machine-readable format, and must include DTP-OSM unique feature IDs, OSM unique feature IDs, and, where applicable, Vicmap feature IDs.

Scenario 3: Handle missing or incomplete exception reports

Given any exceptions are not captured in the report

When the report is reviewed

Then the vendor must identify the cause

And propose corrective actions for review by DTP's geospatial team

Description: Map matching allows for DTP-OSM to include changes in OSM and Vicmap that are no more than 2 weeks old

Feature 10: Timeliness and Currency of DTP-OSM Updates

Ensure that map matching enables DTP-OSM to incorporate recent changes from OSM and Vicmap while meeting processing and manual review time requirements.

Scenario 1: Incorporate recent source changes

Given OSM and Vicmap provide updated road network data

And changes in each source may be up to two weeks old

When the map matching process is executed

Then DTP-OSM must incorporate all changes that are no more than two weeks old

And the resulting dataset must reflect the latest available source geometry and attributes

Scenario 2: Map matching processing time meets requirement

Given the vendor has initiated a complete map matching run

When the algorithm is executed

Then the total processing time must be less than 24 hours

And the vendor must report the start and end time of the processing run

Scenario 3: Handle breaches of timeliness

Given processing time exceeds 24 hours

When the report is reviewed

Then the vendor must identify the cause of the delay

And propose corrective actions for review by DTP's geospatial team

5.4. Fit for Purpose

Description: DTP-OSM has required attributes and geometry for use in LRS

Feature 11: Fit for Purpose of DTP-OSM for LRS

Ensure that DTP-OSM contains all required attributes and geometry for use in the Linear Referencing System (LRS).

Scenario 1: Verify required attributes are present

Given DTP-OSM has been produced from source datasets and custom attributes

When the dataset is reviewed for attribute completeness

Then all attributes required for LRS use must be present

And the vendor must provide a report showing:



- Total number of features reviewed
- Any features missing required attributes
- The names of missing attributes and potential reasons
- The report must be delivered within 24 hours in a structured, machine-readable format, and must include DTP-OSM unique feature IDs, OSM unique feature IDs, and, where applicable, Vicmap feature IDs.

Scenario 2: Verify required geometry is present

Given DTP-OSM includes road geometry from OSM, Vicmap, and custom edits

When the dataset is reviewed for geometric suitability

Then all geometry required for LRS use must be present

And the vendor must provide a report showing:

- Total number of features reviewed
- Any features missing required geometry or connectivity issues
- Potential reasons for missing geometry
- The report must be delivered within 24 hours in a structured, machine-readable format, and must include DTP-OSM unique feature IDs, OSM unique feature IDs, and, where applicable, Vicmap feature IDs.

Scenario 3: Handle missing attributes or geometry

Given any required attributes or geometry are missing

When the report is reviewed

Then the vendor must identify the cause

And propose corrective actions for review by DTP's geospatial team

5.5. Accuracy

Description: Attributes should be matched to correct road segments

Feature 12: Accuracy of Attribute Assignment in DTP-OSM

Ensure that attributes are correctly matched to the corresponding road segments, minimising misassignment from intersecting roads.

Scenario 1: Verify attribute assignment accuracy

Given the DTP-OSM dataset of Priority roads* segments

And intersecting road segments are present in the dataset

When attributes are assigned to each road segment

Then less than 1% of features may have attributes incorrectly matched from intersecting roads

Scenario 2: Reporting of attribute assignment accuracy

Given the attribute assignment review has been completed

When the vendor delivers the DTP-OSM dataset

Then a report must be provided showing:

- Total number of features reviewed
- Number and percentage of features with incorrectly matched attributes
- Any patterns or clusters of misassigned attributes, with potential reasons
- The report must be delivered within 24 hours in a structured, machine-readable format, and must include DTP-OSM unique feature IDs, OSM unique feature IDs, and, where applicable, Vicmap feature IDs.

Scenario 3: Handle incorrect attribute assignments

Given any features are identified with incorrectly matched attributes

When the report is reviewed

Then the vendor must identify the cause of misassignment

And propose corrective actions for review by DTP's geospatial team



Description: Roads should have consistent attributes along the road, with no gaps

Feature 13: Accuracy of Road Attribute Consistency in DTP-OSM

Ensure that road attributes are consistent along the road, allowing roads to be correctly built with minimal segmentation.

Scenario 1: Verify attribute consistency along roads

Given the DTP-OSM dataset of DTP-managed roads

When each road is evaluated for attribute consistency

Then roads should be able to be built for all DTP roads

And fewer than 2 additional features should be created for each road due to attribute gaps

Scenario 2: Reporting of road attribute consistency

Given the road attribute consistency review has been completed

When the vendor delivers the DTP-OSM dataset

Then a report must be provided showing:

- Total number of DTP-managed roads reviewed
- Number of roads with attribute inconsistencies
- Number of additional features created per roads due to gaps
- Any patterns or reasons for inconsistencies
- The report must be delivered within 24 hours in a structured, machine-readable format, and must include DTP-OSM unique feature IDs, OSM unique feature IDs, and, where applicable, Vicmap feature IDs.

Scenario 3: Handle attribute inconsistencies along roads

Given any roads are built with more than 2 additional features due to attribute gaps

When the report is reviewed

Then the vendor must identify the cause of inconsistencies

And propose corrective actions for review by DTP's geospatial team

5.6. Functionality

Description: Enable users to define and update specific geographic areas by drawing a bounding box, ensuring only the selected region is processed while maintaining full network integrity and consistency.

Feature 14: Targeted Area Updates Using Bounding Box Selection

Ensure that users can define a geographic area of interest using a bounding box and update only the selected region while maintaining data integrity, connectivity, and consistency across the full DTP-OSM network.

Scenario 1: User defines a bounding box to select an update area

Given the user is viewing the DTP-OSM network in the Map Manager tool

And the system supports selection of areas using an on-screen bounding box

When the user draws a bounding box around a target area

Then the system must identify all road segments and attributes that fall within or intersect the bounding box

And the selection must be visually confirmed to the user

And the system must prepare the selected features for targeted update processing.

Scenario 2: Targeted updates are processed for the selected area

Given a bounding box-based selection has been created

And the selection contains all relevant geometry, attributes and identifiers

When the user initiates an update for the selected area

Then only the features within the bounding box must be re-processed, refreshed or overwritten



-
- And** the map matching rules must be applied consistently to the selected features
 - And** only the necessary data from OSM, Vicmap and DTP sources must be consumed for this area
 - And** any changes introduced must comply with the required validation rules (fields, domains, defaults, attribute persistence).

Scenario 3: Network integrity is maintained after partial updates

- Given** a partial (bounding-box-based) update has been applied
- When** the updated features are integrated back into the DTP-OSM network
- Then** the full network must remain topologically connected
- And** no new breaks, gaps or attribute inconsistencies may be introduced at the edges of the bounding box
- And** unchanged surrounding features must retain their attributes
- And** all integrity and consistency checks must be re-applied to the updated region and its immediate neighbours.

Appendices

Appendix 1 - 'Priority Roads' Definition

The Priority Road(s) attribute is defined according to each respective dataset. Classifications are noted below:

Custom Attribute (RNDB) –RMA Classification

Priority Roads:

- FW
- AH
- AO

Other:

- MU

Active Roads - OpenStreetMap OSM highway attribute

Priority Roads:

- Motorway, motorway_link,
- Trunk, trunk_link
- Primary, primary_link
- Secondary, secondary_link
- Tertiary, tertiary_link

Other:

- Unclassified
- living_street
- Residential
- Service
- Road
- Footway, pedestrian, path, sidewalk,
- Cycleway
- Track
- construction, proposed

Vicmap – TR Road classification Class code

Priority Roads:

- 0 – 6

Other:

- >= 7



Appendix 2 – Required Attributes

Field Name	Type	Source	Mandatory
id	string	Derived	No
way_id	string	Derived	Yes
tags	json	OSM	Yes
manual_relationship_edit	Boolean	Derived	Yes
vmt_id	string	Derived	Yes
alternative_road_1	string	Vicmap	Yes
alternative_road_2	string	Vicmap	Yes
alternative_road_3	string	Vicmap	Yes
class_code	number	Vicmap	Yes
div_rd	string	Vicmap	Yes
left_locality	string	Vicmap	Yes
local_road	string	Vicmap	Yes
restrictions	string	Vicmap	Yes
right_locality	string	Vicmap	Yes
road_name	string	Vicmap	Yes
road_name_1	string	Vicmap	Yes
road_name_2	string	Vicmap	Yes
road_name_3	string	Vicmap	Yes
road_name_use	string	Vicmap	No
road_seal	string	Vicmap	No
road_status	string	Vicmap	No
road_suffix	string	Vicmap	Yes
road_suffix_1	string	Vicmap	Yes
road_suffix_2	string	Vicmap	Yes
road_suffix_3	string	Vicmap	Yes
road_type	string	Vicmap	Yes
road_type_1	string	Vicmap	Yes
road_type_2	string	Vicmap	Yes
road_type_3	string	Vicmap	Yes
seasonal_close_date	string	Vicmap	No
seasonal_open_date	string	Vicmap	No
ufi	number	Vicmap	No
urban	string	Vicmap	No
vehicular_access	number	Vicmap	No
ARTERIAL	string	RNDB	Yes
AUSLINK	string	RNDB	Yes
CBR	string	RNDB	Yes
CLASSN	string	RNDB	Yes
COMPLEX	string	RNDB	Yes
CWAY	string	RNDB	Yes
DEC_NAME	string	RNDB	Yes



Field Name	Type	Source	Mandatory
DEC_STATUS	string	RNDB	Yes
DEC_TYPE	string	RNDB	Yes
LGA_ABB_NAME	string	RNDB	Yes
LGA_NAME	string	RNDB	Yes
MAINT_ORG	number	RNDB	Yes
MEL30_ART	string	RNDB	Yes
PREV_RMA	number	RNDB	No
PROFILE	number	RNDB	Yes
RAMP	string	RNDB	Yes
RAMP_NAME	string	RNDB	Yes
RAMP_NUM	number	RNDB	Yes
RD_NUM	number	RNDB	Yes
RD_SECTION	number	RNDB	Yes
REGION_NAME	string	RNDB	Yes
RESPONSIBLE_REGION_NAME	string	RNDB	Yes
RMACCLASS	string	RNDB	Yes
RMANUM	number	RNDB	Yes
RMC	number	RNDB	Yes
RNDB_EDIT	string	RNDB	No
RTE_NO	number	RNDB	No
R_CWAY	number	RNDB	No
R_RAMP	number	RNDB	No
R_RTE_NO	number	RNDB	No
SERVICE_RD	string	RNDB	Yes
SRNS1	string	RNDB	Yes
SRNS2	string	RNDB	Yes
SRNS3	string	RNDB	Yes
declared_road	string	Derived	Yes



Appendix 3 - QA and Data Integrity Checks

Below outlines a series of tests that a completed match of OSM / DTP-OSM is subject to. These broadly align to the non-functional specifications for the dataset itself

Sourced from Confluence – page dated 15/8/24

<https://publictransportvic.atlassian.net/wiki/spaces/DEM/pages/3340075210/QA+and+Data+Integrity+Checks>

Table 1 – Validation Processes

Test	Validation Process	Description	Expected Outcome
1	Required Format of the file	Check to ensure that the DTP OSM Road Network file is in required format.	The format of the file should be GeoJSON.
3	Required Format of OSM, VMT, Custom attributes tags	Check that the DTP custom attributes, OSM attributes and VMT attributes are as separate groups in DTP OSM Road Network file.	The import file adheres to the format scripted for MM Export file.
4	Required Fields Exist	Check the field attributes against the DTP OSM Road Network excel file.	All required fields exist and conform to priority attribute list documented and confirmed.
5	Check Field Domains	Check field values against any domains listed in the DTP OSM Road Network excel file.	All field values conform to the domains as listed in the DTP OSM Road Network excel file.
6	Check Class Codes	Check class code field to ensure only class code <= 9 included in the export file.	Class codes values should be <= 9.
8	Road Name Duplication	Check that all VicMap Transport derived road names are unique (i.e. road name and type <> road name 1 and type 1 <> road name 2 and type2)	Unique road name and type for all VicMap Transport derived road names (road name, road name 1, road name 2). Note that Declared name and type is not included in this check.



Test	Validation Process	Description	Expected Outcome
9	Roads specific checks • If RD_NUM <> 9999 then DEC_NAME must be populated • If RD_NUM <> 9999 then RMACCLASS <> MU • If RD_NUM = 9999 then RMACCLASS = MU	Check to ensure that: • All Declared Roads have a Declared Name populated • No Declared Roads have an RMACCLASS of MU • All Local Roads have an RMACCLASS of MU	• All Declared roads have a declared name • No Declared roads have an RMACCLASS of MU • All Local Roads have an RMACCLASS of MU
10	Check Duplicated Records	Check to ensure only 1 single set of attributes is presented for each line segment.	There should be no duplicated records in DTP OSM Road Network import file.
11	Check the existence of unique ID	Check to ensure the unique ID, "id" exists for each road segment as well as OSM Way ID as "parent_id".	There should be one unique ID for each road segment and associated OSM Way ID as parent ID.
12	Check the values of combined road name, type, suffix	Check to ensure the "declared_road", "alternative_road_1", "alternative_road_2" and "alternative_road_3 constructed by concatenating the relevant fields.	declared_road = "Dec_name" + " " + "Dec_type" alternative_road_1 = "road_name_1" + " " + "road_type_1" + " " + "road_suffix_1" alternative_road_2 = "road_name_2" + " " + "road_type_2" + " " + "road_suffix_2" alternative_road_3 = "road_name_3" + " " + "road_type_3" + " " + "road_suffix_3"

Test	Validation Process	Description	Expected Outcome
13	Manual Review of Datasets	Visual check to ensure projection is correct and no obvious errors exist (i.e. features outside Victoria). Ensure that the network maintains proper topological relationships data editing-automation process.	Visual check ok. The network should uphold correct connectivity and spatial relationships
14	Defaults and logic	• Does the ARTERIAL attribute have a default of N? • Does the CLASSN attribute have a default of LO? • Does the MEL30_ART attribute have a default of N? • Does the PREV_RMA attribute have a default of 0? • Does the RD_SECTION attribute have a default of U? • Does the RMACLASS attribute have a default of MU?	Check to see if defaults are applied
15	Declared Road checks	• Do any MU occur on Declared network (is there a polygon to test (ie. Snake&Eggs) • Does the RMANUM attribute have a default of 9999? • Does the RMC attribute have a default of 0? • road_name, road_name_1, road_name_2, road_name_3, have a default of UNNAMED? • MAINT_ORG - is null OK? • If RMACLASS = MU, RTE_NO must Equal 0. • Does the RD_NUM attribute have a default of 9999? • When RMACLASS does not Equal "MU" then RD_NUM <9999. • Do the SRNS1, SRNS2, SRNS3 attributes have a default of N? • Does the RNDB_EDIT attribute have a default of N?	Checks on Declared road network - this is the 15% of the network that has a priority for us.



Test	Validation Process	Description	Expected Outcome
16	LGA and DTP Regions	• LGA_ABB_NAME is populated and valid (check against Polygon) • LGA_NAME is populated and valid (check against Polygon) • REGION_NAME is populated and valid (check against Polygon) • RESPONSIBLE_REGION_NAME is populated (spot check the known 1%)	These line attributes can be checked against other datasets.
17	Profiles	• Does the CBR attribute have a default of N? • Does the CWAY attribute have a default of B? • Does the PROFILE attribute have a default of 1? • Does the RAMP_NUM attribute have a default of 0? • Does the SERVICE_RD attribute have a default of N? • Does the AUSLINK attribute have a default of N? • Is there a DEC_NAME, DEC_STATUS and DEC_TYPE if RMACCLASS DOES NOT EQUAL "MU"?	These attributes will be impacted by the attribute editing process as we are copying these attributes into the wrong area. Should these be defaulted for the unmatched segments so we don't populate with misleading data?

Appendix 4 - Example Results of DTP Automated Review of DTP-OSM

Unmatched Road Segment Analysis — DTP OSM Network 2.3



By Alex Sha 2 min

Version Comparison: Feb 2025 (v1.9) → Apr 2025 (v2.1) → Jun 2025 (v2.2) → Aug 2025 (v2.3)

Scope: Snapshot comparison of **RNDB vs Vicmap** unmatched segments. Priority classes: trunk, motorway, primary, secondary, tertiary (and their `_link` versions).

Overview

This report compares unmatched segments across the **DTP OSM Network** at four critical points:

- **v1.9 – Feb 2025** (25_02_11_dtp_osm_network_osm_tags_ppr) - Total Segment Count: 1,134,409
- **v2.1 – Apr 2025** (25_04_14_dtp_osm_road_network_vic_only) - Total Segment Count: 1,190,169 → **+4.91%** from v1.9
- **v2.2 – Jun 2025** (25_06_10_dtp_osm_road_network_vic_only) - Total Segment Count: 1,190,585 → **+0.03%** from v2.1
- **v2.3 – Aug 2025** (25_08_14_dtp_osm_road_network_vic_only) - Total Segment Count: 1,217,018 → **+2.22%** from v2.2

The analysis focuses on **priority road categories** and tracks unmatched volume trends, percentage change, and system performance over time.

📊 All Roads — Total Unmatched Segments (v1.9 → v2.3)

Dataset	v1.9 (Feb 2025)	v2.1 (Apr 2025)	v2.2 (Jun 2025)	v2.3 (Aug 2025)	Δ v2.3–v2.2	% Change v2.3–v2.2	Δ v2.3–v1.9	% Change v2.3–v1.9
VicMap_Unmatch_Total	308,301	383,969	389,403	379,782	–9,621	–2.47%	+71,481	+23.19%
RNDB_Unmatch_Total	210,797	273,626	299,120	317,515	+18,395	+6.15%	+106,718	+50.63%

📁 Priority Roads – RNDB Unmatched Trend



Road Category	v1.9 (Feb)	v2.1 (Apr)	v2.2 (Jun)	v2.3 (Aug)	Δ v2.2→v2.3	% Change v2.2→v2.3
trunk	1,168	2,537	2,179	2,302	+123	+5.64%
trunk_link	224	310	313	327	+14	+4.47%
motorway	247	432	393	400	+7	+1.78%
motorway_link	497	682	704	707	+3	+0.43%
primary	1,110	3,760	3,066	3,159	+93	+3.03%
primary_link	176	361	398	409	+11	+2.76%
secondary	719	4,109	3,121	3,329	+208	+6.66%
secondary_link	95	198	222	240	+18	+8.11%
tertiary	927	8,364	6,622	7,284	+662	+10.00%
tertiary_link	65	249	285	305	+20	+7.02%
TOTAL	5,228	20,002	17,303	18,462	+1,159	+6.70%

🕒 Tertiary and secondary segments continue to dominate unmatched counts. Small but steady increases across all priority classes.

📊 Priority Roads – Vicmap Unmatched Trend

Road Category	v1.9 (Feb)	v2.1 (Apr)	v2.2 (Jun)	v2.3 (Aug)	Δ v2.2→v2.3	% Change v2.2→v2.3
trunk	1,168	759	832	876	+44	+5.29%
trunk_link	224	284	230	243	+13	+5.65%
motorway	247	162	163	164	+1	+0.61%
motorway_link	497	401	417	419	+2	+0.48%
primary	1,110	1,252	1,434	1,486	+52	+3.63%
primary_link	176	429	312	319	+7	+2.24%

secondary	719	1,074	1,180	1,278	+98	+8.31%
secondary_link	95	270	186	205	+19	+10.22%
tertiary	927	2,060	2,142	2,410	+268	+12.51%
tertiary_link	65	429	238	256	+18	+7.56%
TOTAL	5,228	7,130	7,144	7,656	+512	+7.17%

⚙️ *Unlike previous versions, VicMap's priority classes all show moderate increases — especially in tertiary, secondary, and primary.*

🔑 Key Insights

📊 RNDB Trends:

- **+ 106,718 unmatched segments since v1.9**, largely driven by tertiary and primary segments.
- Priority unmatched in **v2.3 increased by +6.18%** vs v2.2.
- Slight increases seen in trunk, tertiary_link, and secondary.

📊 Vicmap Trends:

- **Vicmap unmatched decreased** for the first time since v1.9.
- **–2.47% total drop** between v2.2 and v2.3, indicating success in resolving non-priority mismatches.
- However, **priority unmatched increased by +7.17%**, suggesting matching improvements did not target the most critical classes.



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